

# Unsupervised Learning Laboratories

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## ***k*-medoids and *c*-means**

You can use external libraries for linear algebra operations but you are expected to write your own algorithms.

### **Exercise 1**

Use the `s3.txt` dataset from the previous lab.

1. Run **your** *k*-means code 20 times with  $k = 15$ . Save the minimum, maximum and average values of the loss.
2. Modify your code such that it uses the *k*-means++ initialization. Perform the same experiment of point 1 and compare the obtained values.
3. Build a scree plot by plotting the best (ie. the lowest) loss out of 20 runs for  $k = 2, \dots, 20$  (with *k*-means++). What is the optimal value of  $k$ ?
4. Modify your code to perform *k*-medoids. Repeat the assignment of point 3 with the new implementation.
5. Modify your code to perform *c*-means. Repeat the assignment of point 3 with the new implementation.